

Emergency Department Staffing Optimization

Problem Overview

We aim to optimize staffing levels in an Emergency Department (ED) to ensure timely patient care while balancing cost and resource utilization. The ED is modeled as a queuing system with stochastic arrivals and service times.

System Components

Triage Levels

Each patient is assigned a triage category with a fixed probability:

$$P(\text{Level } i) = p_i.$$

We consider two levels (Urgent, Non-urgent) :

$$p_2 = 0.40, \quad p_3 = 0.60.$$

Service Times

Service time for a triage level i is assumed exponential with rate μ_i :

$$S_i \sim \text{Exp}(\mu_i), \quad \mathbb{E}[S_i] = \frac{1}{\mu_i}.$$

Example service rates (Urgent, Non-urgent):

$$\mu_2 = 4, \quad \mu_3 = 6.$$

Higher acuity = shorter mean service time but stricter wait limits.

Dynamic Staffing Decisions

Let S_t denote the number of active staff at time t . Then, the controller chooses an action

$$a_t \in \{-1, 0, +1\},$$

representing a request to decrease or increase staff. Staffing is bounded:

$$0 \leq S_t \leq S_{\max}.$$

Increases take effect immediately. Decreases will only be effective when staff finish service; if no staff become free, staffing remains unchanged. The next staffing level is

$$S_{t+1} = S_t + a_t^{\text{effective}}$$

Performance Metrics

Let $C_{i,t}$ denote the number of level- i patients complete treatment during t . During interval t , we have:

- Complete treatment for a level- i patient yields reward $r_2 > r_3 > 0$.
- Maintain S_t staff cost $c_m S_t$.
- Activate new staff cost c_a .
- Block P_t patient receive $c_p P_{\text{ovf},t}$ penalty.

Then, the reward will be:

$$R_t = -c_m S_t - c_a \mathbf{1}_{\text{new activation}} - c_p P_{\text{ovf},t} + \sum_{i=2}^3 r_i C_{i,t}$$

We will calculate this using the simulation above.

Simulation and Optimization

A discrete-event simulation evaluates each policy by reproducing arrivals, queuing, and service. We compare policies via utility U and optionally apply:

- Static approach : Apply the same number of staff for every state.
- Greedy approach: Apply the number of staff based on the number of patients left in the queue.
- Reinforcement Learning approach: Apply MDP algorithm for the problem, using simulation to get rewards.

The goal is to identify staffing rules that maximize U and keep waiting times within triage limits.

Initial simulation environment

Note: Since we incorporate both live simulation in the code, please refer to the Jupyter Notebook for the live simulation

For training, computing and comparing every approaches, we use the following parameters:

Queue Capacity: $M = 20$
Maximum Staff: $S = 10$
Arrival rate: $\mu = 10$
Service rate: $\lambda_2 = 4.0$
 $\lambda_3 = 6.0$
Reward for urgent $r_2 = 50$
Reward for non-urgent $r_3 = 20$
Penalty for Maintenance $c_m = 10$
Penalty for Activate $c_a = 50$
Penalty for blocking $c_p = 100$

This is just the initial parameter. We will change them later for analyze specific properties of each approaches.

Remark

We can easily see that:

Reward system gets + Reward lost = Max reward from arrival patients

where reward lost can be estimated with the number of blocked patients. In other word, if a patient isn't blocked, we guarantee to get the reward from that patient. Therefore, we can ignore rewards from patients and only focus on the lost.

Static Approach

We keep the same number of staffs for every states of the problem.

Optimal solution

By running simulation for different number of staffs, the optimal number of staffs for the above environment is 3. This also align with the theoretical optimal value.

Theoretical Optimal

Important properties

Given n i.d.p random variable $X_1, X_2, \dots, X_n$ where $X_i \sim \text{Exp}(\lambda_i)$, we have:

$$\min(X_1, \dots, X_n) \sim \text{Exp}\left(\sum_{i=1}^n \lambda_i\right) \quad (\text{Proven in class})$$

Therefore, in a system consist of n busy servers with service rate of λ , the transition rate to completeing 1 task is $n\lambda$.

General Double-Class system

Let the number of staff in this policy is s. We define state space as (w_2, w_3, s_2, s_3) where w_2, w_3, s_2, s_3 are the number of urgent, non-urgent patients in queue and number of urgent, non-urgent patients in service. Then, we will have the following transition:

$$\begin{aligned} (w_2, w_3, s_2, s_3) &\xrightarrow{p_2\mu} (w_2 + 1, w_3, s_2, s_3) && (s_2 + s_3 = s, w_2 + w_3 < M) \\ (w_2, w_3, s_2, s_3) &\xrightarrow{p_3\mu} (w_2, w_3 + 1, s_2, s_3) && (s_2 + s_3 = s, w_2 + w_3 < M) \\ (w_2, w_3, s_2, s_3) &\xrightarrow{s_2\lambda_2} (w_2 - 1, w_3, s_2, s_3) && (s_2 + s_3 = s, w_2 > 0) \\ (w_2, w_3, s_2, s_3) &\xrightarrow{s_3\lambda_3} (w_2 - 1, w_3, s_2 + 1, s_3 - 1) && (s_2 + s_3 = s, w_2 > 0, s_3 > 0) \\ (0, w_3, s_2, s_3) &\xrightarrow{s_2\lambda_2} (0, w_3 - 1, s_2 + 1, s_3 - 1) && (s_2 + s_3 = s, w_3 > 0) \\ (0, w_3, s_2, s_3) &\xrightarrow{s_3\lambda_3} (0, w_3 - 1, s_2, s_3) && (s_2 + s_3 = s, w_3 > 0, s_3 > 0) \\ (0, 0, s_2, s_3) &\xrightarrow{p_2\mu} (0, 0, s_2 + 1, s_3) && (s_2 + s_3 < s) \\ (0, 0, s_2, s_3) &\xrightarrow{p_3\mu} (0, 0, s_2, s_3 + 1) && (s_2 + s_3 < s) \\ (0, 0, s_2, s_3) &\xrightarrow{s_2\lambda_2} (0, 0, s_2 - 1, s_3) && (s_2 > 0) \\ (0, 0, s_2, s_3) &\xrightarrow{s_3\lambda_3} (0, 0, s_2, s_3 - 1) && (s_3 > 0) \\ (w_2, w_3, s_2, s_3) &\xrightarrow{\mu} (w_2, w_3, s_2, s_3) && (w_2 + w_3 = M, s_2 + s_3 = s) \end{aligned}$$

Let $P(w_2, w_3, s_2, s_3)$ denote the state probability. Thus, balance equations:

Interior states ($s_2 + s_3 = s, w_2 > 0, w_2 + w_3 < M$):

$$\begin{aligned} P(w_2, w_3, s_2, s_3)(p_2\mu + p_3\mu + s_2\lambda_2 + s_3\lambda_3) &= P(w_2 - 1, w_3, s_2, s_3)p_2\mu \\ &\quad + P(w_2, w_3 - 1, s_2, s_3)p_3\mu \\ &\quad + P(w_2 + 1, w_3, s_2, s_3)s_2\lambda_2 \\ &\quad + P(w_2 + 1, w_3, s_2 - 1, s_3 + 1)s_3\lambda_3 \end{aligned}$$

Full queue state ($s_2 + s_3 = s$, $w_2 + w_3 = M$, $w_2 > 0$):

$$P(w_2, w_3, s_2, s_3)(s_2\lambda_2 + s_3\lambda_3) = P(w_2 + 1, w_3, s_2, s_3) s_2\lambda_2 \\ + P(w_2 + 1, w_3, s_2 - 1, s_3 + 1) s_3\lambda_3$$

Priority boundary state ($s_2 + s_3 = s$, $w_2 = 0$, $w_3 > 0$):

$$P(0, w_3, s_2, s_3)(p_3\mu + s_2\lambda_2 + s_3\lambda_3) = P(0, w_3 - 1, s_2, s_3) p_3\mu \\ + P(1, w_3, s_2, s_3) s_2\lambda_2 \\ + P(1, w_3, s_2 - 1, s_3 + 1) s_3\lambda_3$$

Empty queue, servers not full ($w_2 = 0$, $w_3 = 0$, $0 < s_2 + s_3 < s$):

$$P(0, 0, s_2, s_3)(p_2\mu + p_3\mu + s_2\lambda_2 + s_3\lambda_3) = P(0, 0, s_2 - 1, s_3) p_2\mu \\ + P(0, 0, s_2, s_3 - 1) p_3\mu \\ + P(0, 0, s_2 + 1, s_3) s_2\lambda_2 \\ + P(0, 0, s_2, s_3 + 1) s_3\lambda_3$$

Empty queue, servers full ($w_2 = w_3 = 0$, $s_2 + s_3 = s$):

$$P(0, 0, s_2, s_3)(s_2\lambda_2 + s_3\lambda_3) = P(0, 0, s_2 + 1, s_3) s_2\lambda_2 \\ + P(0, 0, s_2, s_3 + 1) s_3\lambda_3$$

Since the system is too big and might be unsolvable, I will approximate the system of double-class system to a single-class system for easier proof. We can approximate the parameters of this system since the arrival rate is moderate compared to the queue capacity and service rate.

Reduced single-class system

The expected service time of both urgent and non-urgent patients is:

$$\mathbb{E}[S_t] \approx p_2 \frac{1}{\lambda_2} + p_3 \frac{1}{\lambda_3} = 0.4 \cdot \frac{1}{4.0} + 0.6 \cdot \frac{1}{6.0} = 0.1 + 0.1 = 0.2.$$

Therefore, the approximate service rate is:

$$\lambda = \frac{1}{\mathbb{E}[S_t]} \approx 5.$$

The expected reward of both urgent and non-urgent time:

$$E[\text{reward}] = 50 * 0.4 + 20 * 0.6 = 32$$

Therefore, we reduce the problem to the following single-class:

$$\begin{aligned} S &= s, & M &= 20, & K &= S + M = 20 + s, \\ \text{arrival rate } \mu &= 10, & \text{service rate } \lambda &= 5. \end{aligned}$$

Similarly, we define the state space as w, s where w is number of patients in queue and s is number of patients in service. Denote $P(w, s)$ the stationary distribution of those state. We can see that there are no states such that $w > 0$ and $s < \max s$. Therefore, instead of represent (w, s) , we can denoted them using a single variable n where $n = w + s$. Therefore, we have the condition:

$$0 \leq n \leq K$$

Then, we obtain the balance equations:

Interior states ($1 \leq n \leq K - 1$):

$$\mu P(n - 1) = \min(n, s) \lambda P(n).$$

Boundary states:

$$\begin{aligned} \mu P(0) &= \lambda P(1), \\ \mu P(K - 1) &= s \lambda P(K). \end{aligned}$$

We have the relation:

$$P(n) = \frac{\mu}{\min(n, s) \lambda} P(n - 1).$$

We solve this recursion by considering two cases.

If $0 \leq n \leq s$

Since $\min(n, s) = n$, and therefore:

$$\begin{aligned} P(n) &= \frac{\mu}{n \lambda} P(n - 1) \\ \Rightarrow P(n) &= \frac{\mu^n}{n! \lambda^n} P(0), \quad 0 \leq n \leq s. \end{aligned}$$

If $s < n \leq K$

Since $\min(n, s) = s$, and therefore:

$$P(n) = \frac{\mu}{s \lambda} P(n - 1)$$

With Case 1, we also have:

$$P(n) = \frac{\mu^n}{s! s^{n-s} \lambda^n} P(0), \quad s < n \leq K$$

Combine this, we have:

$$P(n) = \begin{cases} \frac{\mu^n}{n! \lambda^n} P(0), & 0 \leq n \leq s \\ \frac{\mu^n}{s! s^{n-s} \lambda^n} P(0), & s < n \leq K \end{cases}$$

Normalize the sum to 1, we have:

$$\frac{1}{P(0)} = \sum_{n=0}^s \frac{\mu^n}{n! \lambda^n} + \sum_{n=s+1}^K \frac{\mu^n}{s! s^{n-s} \lambda^n}.$$

The blocking probability is:

$$P_{\text{block}} = P(K) = \frac{\mu^K}{s! s^{K-s} \lambda^K} P(0)$$

Then the rate of entering system is:

$$\mu_{\text{eff}} = \mu(1 - P_{\text{block}})$$

The expected reward with staff level s is:

$$\begin{aligned} R(s) &= E[\text{reward}] \cdot \mu_{\text{eff}} - c_m \cdot s - c_p \cdot \mu P_{\text{block}} \\ &= 32 \cdot \mu(1 - P_{\text{block}}(s)) - 10s - 100 \cdot \mu P_{\text{block}}(s). \end{aligned}$$

Consider s from 1 to 10, we have:

$$\arg \max_{1 \leq s \leq 10} R(s) = 3$$

Greedy Approach

The idea of this approach is to dynamically change the number of staff accordingly to the number of patients in the queue. The policy for this is as follow. Let P_q the number of patient in queue.

- Increase when the queue is likely to be overloaded ($P_q \geq v_o$)
- Decrease when the queue is free ($P_q \leq v_f$)

The threshold will be chosen to:

- Minimizing the cost of maintenance (v_f)

- Minimizing the penalty of blocking (v_o)

Also, by only increase and decrease the number of staffs when needed, it will also minimize the cost of increasing a staff.

Theoretical Idea

The static approach gives us a fixed staff policy so that the expected rewards maximize, and the states of the system will try to converge to a certain set of states. However, what if the states of the system is currently very far from the set of balance states that the optimal static policy want the system to converge to ? Generally, the idea is to increase the staff so that the speed of converging is much faster than the current speed. Therefore, we will try to provide a better policy based on the current optimal static policy, by increase or decrease the current optimal policy by 1 in some occasions.

Let $B = E[\text{Blocked patients at an interval} \mid s, \text{queue size}]$.

Since we have the ability to increase the number of patient, we can decide the either increase or keep the same number of staff based on comparing the expected lost:

$$B(s = s^*) * c_p \text{ vs } B(s = s^* + 1) * c_p + c_a$$

We try to approximate B by getting the blocked probabilities a similar system:

$$\mu = 10, \lambda = 5, S = s, M = a(\text{number of slots in queue left}), K = a + s$$

Recall from above:

$$P_{\text{block}} = \frac{\mu^K}{s! s^{K-s} \lambda^K} P(0)$$

where

$$\frac{1}{P(0)} = \sum_{n=0}^s \frac{\mu^n}{n! \lambda^n} + \sum_{n=s+1}^K \frac{\mu^n}{s! s^{n-s} \lambda^n}.$$

$$\Rightarrow B = P_{\text{block}} * \mu$$

Therefore, we find the threshold such that:

$$(P_{\text{block}, s^*} - P_{\text{block}, s^*+1}) * \mu * c_p < c_a$$

Apply every a from 0 to 20, we have the biggest a as:

$$\max a = 2$$

Therefore, we choose the upper threshold as 18.

For lower threshold, we choose 16 as the value.

As we test in the simulation, we realize that the greedy approach isn't much better than the static approach, perhaps since it is only activated in some specific cases, that the probability of entering that cases is minimal.

Reinforcement Learning Approach

In this approach, we model the problem as a MDP, and use Q-learning algorithm to train the model and find the optimal policy.

RL Environment

State Space:

$$S = (Q_2, Q_3, B_2, B_3, S_t)$$

where

- Q_2, Q_3 the number of urgent and non-urgent patients in queue.
- B_2, B_3 the number of urgent and non-urgent patients in service.
- S_t current number of staff

Action Space:

$$A = \{-1, 0, 1\}$$

represent the change of staff after that interval.

Reward Function:

Base on the simulation provided above, we can simulate the system in an interval and return the reward obtained from that interval.

Performance Comparison

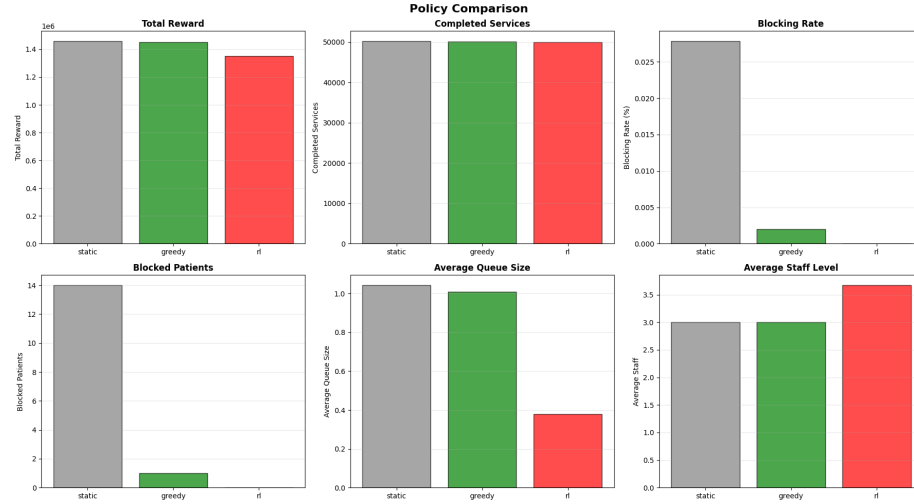


Figure 1: Performance comparison

From the above figure, we can see 3 unexpected results: RL is the lowest, static and greedy don't have much different. This can be explained by the following reasons:

1. The state space of RL is too big, and it grow quadratically . Recall that $M = 20, S = 10$, then the state space (Q_1, Q_2, B_1, B_2, S) has more than 50000 states. Therefore, the model isn't sufficient to accomplish this task.
2. The parameter favors stability, where static approach has the advantage, while greedy algorithm doesn't provide much impact

If we change the parameters to $\mu_2 = 2, \mu_3 = 3, c_a = 5, c_p = 50$

$$\Rightarrow s^* = 6$$

and we set the threshold accordingly, then we have the following, giving the similar properties.

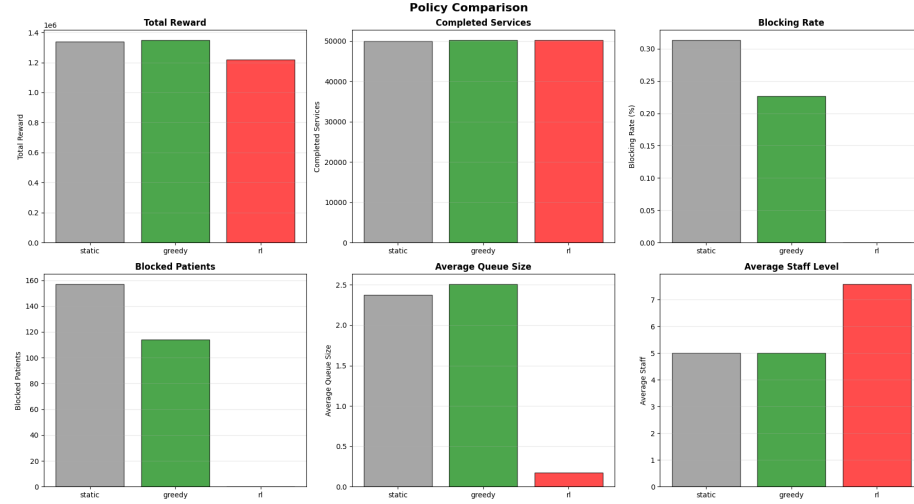


Figure 2: Performance comparison